$\frac{1}{\theta_y}$ $x \in [0, \theta_y)$

$(\ )(\ )$

$n \times l$ $l \times m$

$\theta_y = e^4$ $\theta_z = e^{10}$

$n \times m$

$$D_{KL} = \int_0^{e^4} \frac{1}{\theta_y} \ln\left(\frac{\frac{1}{\theta_y}}{\frac{1}{\theta_z}}\right) dx$$

$$= \int_0^{e^4} e^{-4} \ln\left(\frac{e^{-4}}{e^{-10}}\right) dx$$

$$= e^{-4} \int_0^{e^4} (\ln(e^{-4}) - \ln(e^{-10})) dx$$

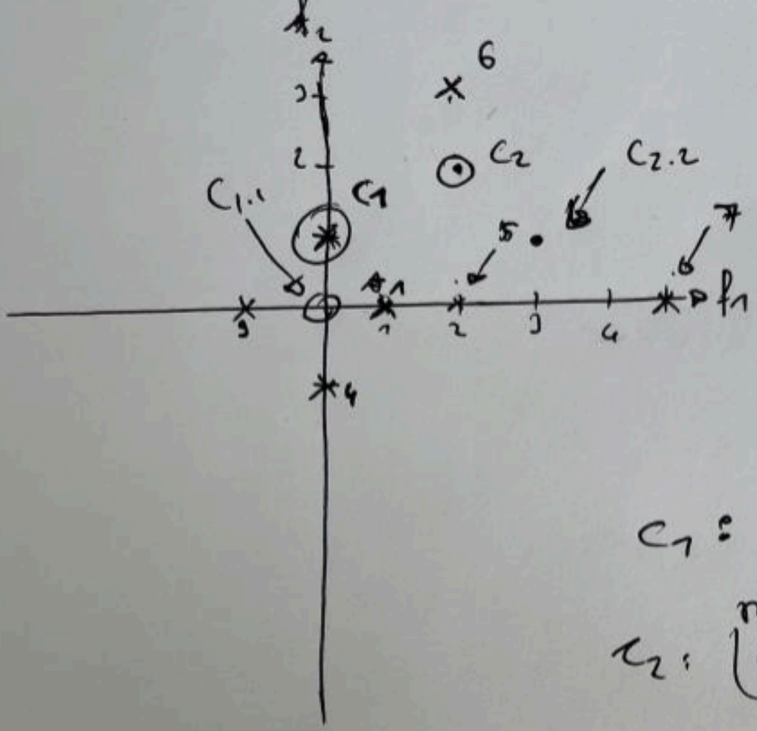
$$= e^{-4}\left(\int_0^{e^4} -4 \, dx - \int_0^{e^4} -10 \, dx\right)$$

$$= e^{-4}\left(-4 \cdot e^4 + 10 \cdot e^4\right)$$

$$= e^{-4} \, 6e^4 = 6$$

(10)

$\mu = c_1 \quad v = c_2$



c_1 : (1 - 4) points

c_2 : (5 - 7) points

$c_1 =$ middle of
$(1,0), (0,1), (-1,0)\ (0,-1)$
$= (0,0)$

$c_2 =$ ~~[illegible]~~ mean of
$(2,0)$
$+ (2,3)$
$+ (5,0)$

$\frac{9,3}{3} = (3,1)$

Factor Analysis: Which statements are correct?

We keep the notation from the lecture:

n is the number data samples

m is the dimension of each data sample

l is the number of factors

$\mathbf{y}$ is the factor vector

$\mathbf{U}$ is the factor loading matrix

$\mathbf{\Psi}$ is the noise covariance matrix

ϵ is the noise vector

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- [ ] a. $\mathbf{\Psi}$ has shape $m \times m$.
- [ ] b. $\mathbf{\Psi}$ has shape $l \times l$.
- [x] c. $\mathbf{\Psi}$ has shape $n \times n$. ✗
- [x] d. $\mathbf{U}$ has shape $m \times l$. ✓
- [ ] e. $\mathbf{U}$ has shape $m \times n$.
- [ ] f. $\mathbf{U}$ has shape $l \times m$.
- [ ] g. ϵ has dimension m.
- [ ] h. ϵ has dimension l.
- [x] i. ϵ has dimension n. ✗
- [ ] j. $\mathbf{y}$ has dimension m.
- [x] k. $\mathbf{y}$ has dimension l. ✓
- [ ] l. $\mathbf{y}$ has dimension n.

- [x] k. $\mathbf{y}$ has dimension l.
- [ ] l. $\mathbf{y}$ has dimension n.

Frage 4
Falsch
Erreichte Punkte 0,00 von 5,00
Frage markieren

Factor Analysis:

You are given the factor loading matrx $\mathbf{U}$ and the noise covariance matrix $\mathbf{\Psi}$:

$$\mathbf{U} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$$

$$\mathbf{\Psi} = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

Compute the first row of the covariance matrix $\mathbf{C}$ and enter its (integer) values below.

- $C_{11} =$ 30 ✖
- $C_{12} =$ 0 ✖
- $C_{13} =$ 0 ✖

Frage 5
Teilweise richtig
Erreichte Punkte 1,25 von 5,00
Frage markieren

Which statements about Scaling and Projection Methods are correct?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- [x] a. Isomap uses a geodesic distance matrix, whose eigenvectors are the coordinates of the projected space. ✔
- [ ] b. The objective in t-SNE is minimization of the KL divergence between the similarities of the (high-dimensional) inputs and the (low-dimensional) mapped outputs.
- [x] c. In locally linear embedding (LLE), each data point is described as a linear combination of its neighbors. ✔
- [ ] d. The objective of scaling is to project data into a higher-dimensional space, e.g. to allow better model selection.
- [ ] e. In self-organizing maps the objective is to minimize an energy (error) function.
- [ ] f. Multidimensional scaling (MDS) keeps the distances between data points (as well as possible).
- [ ] g. LLE in general performs linear mappings to realize neighborhood-preserving embeddings.

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- [ ] e. In self-organizing maps the objective is to minimize an energy (error) function.
- [ ] f. Multidimensional scaling (MDS) keeps the distances between data points (as well as possible).
- [ ] g. LLE in general performs linear mappings to realize neighborhood-preserving embeddings.
- [x] h. In t-SNE, the similarity between data points i and j is given by their joint probability. ✖

Assume you have two random variables Y and Z, which are uniformly distributed on $[0, \theta_Y]$ and $[0, \theta_Z]$, respectively, i.e.

$$p_Y(x) = \begin{cases} \frac{1}{\theta_Y} & \text{if } x \in [0, \theta_Y] \\ 0 & \text{else.} \end{cases}$$

$$p_Z(x) = \begin{cases} \frac{1}{\theta_Z} & \text{if } x \in [0, \theta_Z] \\ 0 & \text{else.} \end{cases}$$

For $\theta_Y = e^4$ and $\theta_Z = e^{10}$, compute the KL divergence

$$D_{\mathrm{KL}}(p_Y || p_Z) = \int_{-\infty}^{\infty} p_Y(x) \; \log[p_Y(x)/p_Z(x)] \, \mathrm{d}x.$$

As for "log", use the natural logarithm.

You can use a pocket calculator. Round to two decimal places. Number format example: 0.67

- $D_{\mathrm{KL}}(p_Y || p_Z) =$ 0,11 ✖ .

Which statements about Non-negative Matrix Factorization (NMF) are correct?

As in the lecture, we have:

$\mathbf{X} = \mathbf{Y}\,\mathbf{U}^T$

$\mathbf{Y} \in \mathbb{R}^{n \times l}$

$\mathbf{U} \in \mathbb{R}^{m \times l}$

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- [ ] a. NMF objectives are the KL divergence and the Euclidean distance.
- [ ] b. NMF is always exactly solvable.
- [x] c. $\mathbf{X} \in \mathbb{R}^{m \times n}$. ✖
- [x] d. The non-negativity constraints make the representation of the observations purely additive. ✔
- [x] e. The the outer product representation reads $\mathbf{X} = \sum_{k=1}^{l} \mathbf{y}_k \mathbf{u}_k^T$. ✔
- [ ] f. NMF factorizes any positive or negative definite matrix into two positive definite matrices.

Which statements about Clustering are correct?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- [x] a. In affinity propagation, the cluster centers are always data points. ✔
- [ ] b. Clustering (or Cluster Analysis) is both a regression and a classification technique.
- [ ] c. In both k-means and fuzzy k-means, the membership distribution is always given by a binary random variable.
- [x] d. In mixture models, an observation is assigned to the component with the largest prior distribution. ✖
- [ ] e. Top-down clustering relies on building the minimal spanning tree and then in each step removes the largest of the remaining edges.
- [ ] f. In a mixture of Gaussians, the maximum likelihood and the maximum posterior always have the same value.
- [ ] g. The agglomerative hierachical clustering algorithm repeatedly fuses the two maximally distant clusters.
- [x] h. Similarity-based clustering does not require to represent objects via feature vectors. ✔

Mixture model:

Assume the following case:

The prior probability of component j is 0.1.

The probability for observing a data sample $\mathbf{x}$ given the total parameter set $\boldsymbol{\theta}$ is 0.2.

The likelihood for observing a data sample $\mathbf{x}$, given component j and corresponding parameter vector, is 0.5.

Compute the posterior probability of component j.

You can use a pocket calculator. State your final result with two decimal places. Number format example: 0.67

- Posterior = 0,04 ✖

Assume you want to do k-means on the following 2-dimensional dataset:

$X = \{(1,0), (0,1), (-1,0), (0,-1), (2,0), (2,3), (5,0)\}$.

The initial cluster centers are $\boldsymbol{\mu} = (0,1)$ and $\boldsymbol{\nu} = (2,2)$.

Compute the coordinates of the new cluster centers $\boldsymbol{\mu}^{\text{new}} = (\mu_1^{\text{new}}, \mu_2^{\text{new}})$ and $\boldsymbol{\nu}^{\text{new}} = (\nu_1^{\text{new}}, \nu_2^{\text{new}})$ after one interation of k-means.

Hint: Make a sketch first. The problem can mainly be solved graphically.

- $\mu_1^{\text{new}} =$ 0 , $\mu_2^{\text{new}} =$ 0
- $\nu_1^{\text{new}} =$ 3 , $\nu_2^{\text{new}} =$ 1

Which statements about Biclustering are correct?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- [x] a. FABIA is a generative model. ✔
- [ ] b. FABIA defines biclusters as inner products.
- [x] c. Row and column elements must belong to exactly one bicluster or no bicluster at all. ✘
- [ ] d. In FABIA, the likelihood is analytically intractable and requires, e.g., the (variational) EM algorithm.
- [ ] e. A common biclustering method is the varicance maximization method.
- [x] f. Biclustering simultaneously clusters the rows and columns of a matrix. ✔

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$$\mathbf{\Psi} = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

Compute the first row of the covariance matrix $\mathbf{C}$ and enter its (integer) values below.

- $C_{11} =$ 10
- $C_{12} =$ 11
- $C_{13} =$ 17